

Abhiuday Narayan

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EDUCATION

National Institute of Technology, Silchar

B.Tech in Electrical Engineering | CGPA: 7.83/10

Class XII: 91.8%

Silchar, Assam

Expected Graduation: 2028

TECHNICAL SKILLS

Languages: Python, C++, C, SQL, JavaScript

Backend & DevOps: FastAPI (Async), Async SQLAlchemy, MySQL, REST APIs, Ubuntu Server, GitHub CI/CD, Cloudflare Tunnels, JWT Authentication

Data Science & ML: Pandas, NumPy, Scikit-learn, PyTorch, LangChain, LangGraph, OpenCV

Edge AI & Hardware: Raspberry Pi 4, ArduPilot, MAVLink, LoRa, Fusion 360, ONNX Runtime Web, NCNN, TensorFlow Lite, Weights & Biases

Frontend: React, Vite, Tailwind CSS, WebGL, Browser WASM

EXPERIENCE

Machine Learning Research Intern – Drone Lab

National Institute of Technology, Silchar

Feb 2026 – Aug 2026

Silchar, Assam

- Designed a domain-adapted YOLO model with custom CBAM (Channel and Spatial Attention) modules to optimize an early-warning aerial forest fire detection system, prioritizing high recall to minimize catastrophic false negatives.
- Diagnosed and resolved a complete training collapse (mAP dropping to 0) caused by gradient explosion from randomly initialized attention layers; implemented a two-stage warm-up strategy by freezing the COCO-pretrained backbone for 15 epochs to stabilize fine-tuning.
- Deployed the lightweight model on a Raspberry Pi 4 (8GB) utilizing the NCNN framework for ARM-based hardware optimization; deliberately capped inference at 6 FPS to preserve 50% CPU overhead for concurrent telemetry and flight control tasks.

PROJECTS

NIDAR – Edge AI Search-and-Rescue Drone System | *YOLO, NCNN FP16, Raspberry Pi 4, Picamera2, OpenCV, ArduPilot, MAVLink, LoRa*

- Developed a Raspberry Pi-based edge vision pipeline for human detection using NCNN FP16 inference, capturing real-time video via Picamera2 and OpenCV.
- Built a MAVLink telemetry interface to connect with ArduPilot, dynamically logging visual YOLO detections alongside the drone's active GPS position, altitude, and attitude data.
- Architected a dual-band communication system utilizing LoRa for low-latency telemetry and emergency compressed-image fallback, alongside long-range Wi-Fi for primary video transmission.

Edgenix.dev – Client-Side ML Inference Platform | *React, Vite, FastAPI, MySQL, ONNX Runtime Web, WebGL, WebAssembly, GitHub CI/CD*

- Architected a client-side inference platform hosting 7 lightweight YOLO Nano models (including fire, human, and pothole detection) using ONNX Runtime Web and WebGL, offloading compute to user devices to eliminate server-side GPU costs.
- Deployed an asynchronous FastAPI backend on a self-managed Ubuntu Server homelab, engineering a GitHub CI/CD pipeline for automated updates and utilizing Cloudflare Tunnels to securely expose the local server through restrictive university network firewalls.
- Implemented secure user authentication and model distribution using MySQL and JWTs, managing state and UI rendering via React and Vite.

ACHIEVEMENTS & LEADERSHIP

- 2nd Place – Robotron Heavyweight Robowar (15 teams): Designed and fabricated a heavyweight combat robot using Fusion 360.
- Top 7 among 150+ teams – Neurathon '26: Developed a multi-agent event logistics platform utilizing LangChain, LangGraph, and the OpenAI API for automated task coordination.
- 1st Place – Electra Society Hackathon: Conceptualized and pitched a comprehensive hardware/software disaster management platform.
- Executive Member – N.E.R.D.S Robotics Club: Mentor junior members in robotics fundamentals and organize technical workshops.